

Policy Gradient Methods

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Reinforcement Learning Summer School 2026

June 5, 2026

Outline

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Policy Gradient Method

Natural Policy Gradient & TRPO

Entropy Regularization in Policy Optimization

Basics of Policy Optimization

Basic Setting: Discounted-Reward MDPs

An infinite-horizon, discounted-reward MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu)$, where

- ▶ $\mathcal{S}$ is a countable state space and $\mathcal{A}$ is a finite action space
- ▶ $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta_{\mathcal{S}}$ is the transition function; $P(s'|s, a)$
- ▶ $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward,
- ▶ $\gamma \in [0, 1)$ is the discount factor,

Under a given policy $\pi : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$, we define

$$V^{\pi}(s) := \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s \right]$$

$$Q^{\pi}(s, a) := \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a \right]$$

under the dynamics

$$s_{t+1} \sim P(\cdot | s_t, a_t) \quad \text{and} \quad a_t \sim \pi(\cdot | s_t).$$

Policy Optimization Problem

For a given initial state distribution $\mu \in \Delta_{\mathcal{S}}$, let

$$J^{\pi}(\mu) := \mathbb{E}_{s_0 \sim \mu}[V^{\pi}(s_0)] = \sum_{s \in \mathcal{S}} \mu(s) V^{\pi}(s).$$

Given a **differentiable** class of randomized policies

$$\Pi_{\Theta} := \{\pi_{\theta} : \theta \in \Theta \subset \mathbb{R}^p\},$$

the policy optimization aims to solve

$$\max_{\theta \in \Theta} J^{\pi_{\theta}}(\mu) \quad (\text{PO})$$

Example: Tabular SoftMax Policies

$$\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})}, \quad \theta \in \Theta = \mathbb{R}^{|\mathcal{S} \times \mathcal{A}|}$$

Challenges in Policy Optimization

- **Gradient computation:** For gradient ascent

$$\theta_{k+1} = \theta_k + \eta_k \nabla_{\theta} J^{\pi_{\theta_k}}(\mu), \quad k \in \mathbb{N},$$

estimate $\nabla_{\theta} J^{\pi_{\theta}}(\mu)$ *without* using P or r .

Solution: Policy gradient theorem (next)

- **Large state spaces:** In general, $|\mathcal{S}|$ can be large or infinity.

Solution: Function approximation

$$\pi_{\theta}(a|s) = \frac{e^{f_{\theta}(s,a)}}{\sum_{a' \in \mathcal{A}} e^{f_{\theta}(s,a')}}, \quad \theta \in \Theta \subset \mathbb{R}^d$$

where $d \ll |\mathcal{S} \times \mathcal{A}|$.

Examples: Π_{Θ} with Function Approximation

1. Linear: Given a feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$,

$$f_{\theta}(s, a) = \theta^{\top} \phi(s, a).$$

2. Neural: $f_{\theta}(s, a) = \sum_{i=1}^m c_i \sigma(w_i^{\top} \varphi(s, a))$ with $\theta = (c_i, w_i)_{i \in [m]}$.

Policy Gradient Method

Policy Gradient Theorem I: REINFORCE

Theorem (Policy Gradient Theorem [Wil92])

Under $\pi_\theta \in \Pi_\Theta$, let $s_0 \sim \mu$,

$$\tau := (s_0, a_0, s_1, a_1, \dots) \quad \text{and} \quad R(\tau) := \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t).$$

Then, $\tau \sim p_\theta$ with $p_\theta(\tau) = \mu(s_0) \prod_{t=0}^{\infty} \pi_\theta(a_t|s_t) P(s_{t+1}|s_t, a_t)$, and

$$\nabla_\theta J^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim p_\theta} \left[R(\tau) \sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t|s_t) \right].$$

Remark (Model-free policy optimization via REINFORCE)

Unbiased and consistent stochastic estimate of $\nabla_\theta J^{\pi_\theta}(\mu)$ without using P or r :

$$\tau^{(i)} \stackrel{\text{iid}}{\sim} p_\theta \quad \rightsquigarrow \quad \widehat{\nabla}_\theta J^{\pi_\theta}(\mu) = \frac{1}{n} \sum_{i=1}^n R(\tau^{(i)}) \sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t^{(i)}|s_t^{(i)})$$

Then, $\mathbb{E}[\widehat{\nabla}_\theta J^{\pi_\theta}(\mu)] = \nabla_\theta J^{\pi_\theta}(\mu)$ and $\widehat{\nabla}_\theta J^{\pi_\theta}(\mu) \xrightarrow{n \rightarrow \infty} \nabla_\theta J^{\pi_\theta}(\mu)$ a.s.

Policy Gradient Theorem I: REINFORCE

Proof of Policy Gradient Theorem I.

Let

$$F(\theta) = \mathbb{E}_{v \sim p_\theta} [g(v)] = \int g(v) p_\theta(v) dv,$$

where $g : \mathcal{V} \rightarrow \mathbb{R}$ and p_θ is a density function. Then

$$\begin{aligned} \nabla_\theta F(\theta) &= \int g(v) \nabla_\theta p_\theta(v) dv \\ &= \int g(v) p_\theta(v) \nabla_\theta \log p_\theta(v) dv = \mathbb{E}_{v \sim p_\theta} [g(v) \nabla_\theta \log p_\theta(v)]. \end{aligned}$$

In our case,

$$\nabla_\theta \log \left(\mu(s_0) \prod_{t=0}^{\infty} \pi_\theta(a_t | s_t) P(s_{t+1} | s_t, a_t) \right) = \sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t | s_t).$$



Policy Gradient: Beyond REINFORCE

Theorem (Policy Gradient Theorem II)

$$\nabla_{\theta} J^{\pi_{\theta}}(\mu) = \mathbb{E}_{\tau \sim p_{\theta}} \left[\sum_t \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \mid s_0 \sim \mu \right].$$

► Let $d_{\mu}^{\pi}(s) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s | s_0 \sim \mu)$.

Corollary (Policy Gradient Theorem III)

$$\nabla_{\theta} J^{\pi_{\theta}}(\mu) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi}, a \sim \pi_{\theta}(\cdot | s)} [Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)].$$

It is easy to sample from d_{μ}^{π} [AKLM21]:

$$T \sim \text{Geo}(1 - \gamma) \quad \Rightarrow \quad s_T \sim d_{\mu}^{\pi}$$

Then, it is easy to estimate Q^{π} :

$$H \sim \text{Geo}(1 - \gamma) \quad \Rightarrow \quad \hat{Q}^{\pi}(s_T, a_T) = \sum_{k=0}^H r(s_{T+k}, a_{T+k})$$

An unbiased estimator of $\nabla J^{\pi_{\theta}}(\mu)$ is $\frac{1}{1 - \gamma} \nabla_{\theta} \log \pi_{\theta}(a_T | s_T) \hat{Q}^{\pi_{\theta}}(s_T, a_T)$.

Policy Gradient: Beyond REINFORCE

Proof ideas: Recall that

$$V^{\pi_\theta}(s) = \sum_a \pi_\theta(a|s) Q^{\pi_\theta}(s, a)$$

$$Q^{\pi_\theta}(s, a) = r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^{\pi_\theta}(s')$$

Step 1: $\nabla V^{\pi_\theta}(s) = \sum_a \nabla \pi_\theta(a|s) Q^{\pi_\theta}(s, a) + \sum_a \pi_\theta(a|s) \nabla Q^{\pi_\theta}(s, a)$

Step 2: $\nabla Q^{\pi_\theta}(s, a) = \gamma \sum_{s'} P(s'|s, a) \nabla V^{\pi_\theta}(s')$

Then,

$$\nabla V^{\pi_\theta}(s) = \mathbb{E}_{a_0} [\nabla \log \pi_\theta(a_0|s) Q^{\pi_\theta}(s, a_0)] + \gamma \mathbb{E}[\nabla V^{\pi_\theta}(s_1) | s_0 = s]$$

For the second part, for any $h : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$,

$$\begin{aligned} \mathbb{E}^{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t h(s_t, a_t) | s_0 \sim \mu \right] &= \sum_{s,a} \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s, a_t = a | s_0 \sim \mu) h(s, a) \\ &= \sum_{s,a} h(s, a) \pi_\theta(a|s) \boxed{\sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s | s_0 \sim \mu)} \end{aligned}$$

Policy Gradient: Baselines

For any (measurable) baseline $b : \mathcal{S} \rightarrow \mathbb{R}$,

$$\nabla_{\theta} J^{\pi_{\theta}}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}, a \sim \pi_{\theta}(\cdot|s)} \left[(Q^{\pi_{\theta}}(s, a) - b(s)) \nabla_{\theta} \log \pi_{\theta}(a|s) \right]$$

Important point is that b is independent of $a \in \mathcal{A}$:

$$\sum_s \sum_a b(s) \underbrace{\pi_{\theta}(a|s) \nabla \log \pi_{\theta}(a|s)}_{=\nabla \pi_{\theta}(a|s)} = \sum_s b(s) \nabla_{\theta} \sum_a \pi_{\theta}(a|s) = 0$$

A standard baseline choice is $b(s) = V^{\pi_{\theta}}(s)$.

Corollary (PG with Advantage – PG-A)

$$\nabla_{\theta} J^{\pi_{\theta}}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}, a \sim \pi_{\theta}(\cdot|s)} \left[A^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a|s) \right],$$

where

$$A^{\pi_{\theta}}(s, a) = Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s).$$

Off-Policy REINFORCE via Importance Sampling

- ▶ **Off-policy sampling.**

- ▶ We want to optimize π_θ ,
- ▶ Trajectory $\tau = (s_0, a_0, \dots)$ comes from a **behavior policy** π_b :

$$p_b(\tau) = \mu(s_0) \prod_{t=0}^{\infty} \pi_b(a_t | s_t) P(s_{t+1} | s_t, a_t).$$

- ▶ **Idea.** Importance sampling [MPK01, DWS12]:

$$\nabla J^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim p_b} \left[\frac{p_\theta(\tau)}{p_b(\tau)} R(\tau) \sum_{t=0}^{\infty} \nabla_\theta \log_\theta(a_t | s_t) \right]$$

- ▶ **Question.** Is IS possible without knowing P ? Yes.

$$\frac{p_\theta(\tau)}{p_b(\tau)} = \frac{\cancel{\mu(s_0)} \prod_{t=0}^{\infty} \pi_\theta(a_t | s_t) \cancel{P(s_{t+1} | s_t, a_t)}}{\cancel{\mu(s_0)} \prod_{t=0}^{\infty} \pi_b(a_t | s_t) \cancel{P(s_{t+1} | s_t, a_t)}} = \prod_{t=0}^{\infty} \frac{\pi_\theta(a_t | s_t)}{\pi_b(a_t | s_t)}$$

- ▶ **Good and bad.**

- ▶ Sample reuse and data efficiency 👍
- ▶ Larger variance in gradient estimate 👎

Nonconcavity of Policy Optimization

PG $\Rightarrow$ unbiased estimates for $\nabla J^{\pi_{\theta}}(\mu)$. Thus,

$$\theta_{k+1} = \theta_k + \eta \nabla_{\theta} J^{\pi_{\theta}}(\mu).$$

Question: CONVEX OPTIMIZATION to conclude that

$$\lim_{k \rightarrow \infty} J^{\pi_{\theta_k}}(\mu) = \sup_{\theta \in \Theta} J^{\pi_{\theta}}(\mu)?$$



Proposition (Nonconcavity of (PO) under SoftMax)

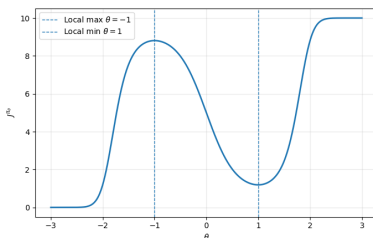
There exists an MDP such that (PO) is nonconcave under SoftMax.

Consider $\mathcal{S} = \{s\}$ and $\mathcal{A} = \{a^{(1)}, a^{(2)}\}$ with

$$r(s, a) = \begin{cases} 1, & a = a^{(1)} \\ 0, & a = a^{(2)} \end{cases} \quad \text{and} \quad \pi_{\theta}(a|s) = \begin{cases} \frac{e^{\theta^3 - 3\theta}}{e^{\theta^3 - 3\theta} + 1}, & a = a_1, \\ \frac{1}{e^{\theta^3 - 3\theta} + 1}, & a = a_2. \end{cases}$$

Then, $J^{\pi_{\theta}}$ is nonconcave.

Nonconcavity of Policy Optimization



Proposition (Nonconcavity of (PO) under SoftMax)

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Then, J^{π_θ} is nonconcave.

Convergence of Policy Gradient

Theorem (Convergence of PG – Exact, [AJKS19])

Assume that $h(\theta) := J^{\pi_{\theta}}(\mu)$ is L -smooth:

$$|h(\theta) - h(\theta_0) - \nabla h(\theta_0)^{\top}(\theta - \theta_0)| \leq \frac{L}{2} \|\theta - \theta_0\|_2^2.$$

Then, exact PG with $\eta = \frac{1}{L}$ yields

$$\min_{t=0,1,\dots,T-1} \|\nabla_{\theta} J^{\pi_{\theta_t}}(\mu)\|_2^2 \leq \frac{2L [J^{\pi_{\theta^*}}(\mu) - J^{\pi_{\theta_0}}(\mu)]}{T}$$

Proof.

$$\begin{aligned} h(\theta_{t+1}) &= h(\theta_t + \eta \cdot \nabla h(\theta_t)) \\ &\geq h(\theta_t) + \eta \cdot \nabla^{\top} h(\theta_t) \nabla h(\theta_t) - \frac{\eta^2 L}{2} \cdot \|\nabla h(\theta_t)\|_2^2 \\ &= h(\theta_t) + \left(\eta - \frac{\eta^2 L}{2} \right) \|\nabla h(\theta_t)\|_2^2. \end{aligned}$$

Telescoping sum over $t = 0, 1, \dots, T-1$ and rearranging concludes proof. $\square$

Need for Exploration

Consider tabular SoftMax: $\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})}$. Then, PG iterates are

$$\theta_{s,a}(k+1) = \theta_{s,a}(k) + \underbrace{\frac{\eta}{1-\gamma} d_{\mu}^{\pi_{\theta(k)}}(s) \pi_{\theta(k)}(a|s)}_{=: \eta_k(s,a)} A^{\pi_{\theta(k)}}(s,a)$$

- ▶ $\eta_k(s,a)$ acts like an effective step-size.
- ▶ If (s,a) is rarely visited, then $\eta_k(s,a)$ is small and $\theta_{s,a}$ is rarely updated.

Natural Policy Gradient & TRPO

Geometry of Policy Optimization and Natural Gradient

PG can be written as

$$\theta_{k+1} \in \arg \max_{\theta \in \mathbb{R}^d} \left\{ \nabla_{\theta} J^{\pi_{\theta}}(\mu)^{\top} (\theta - \theta_k) - \frac{1}{2\eta} \|\theta - \theta_k\|_2^2 \right\}.$$

- ▶ We want steepest ascent in policy space $\{\pi_{\theta} : \theta \in \Theta\}$.
- ▶ Alternatively, consider

$$\theta_{k+1} \in \theta_k + \arg \max_{\Delta \theta \in \mathbb{R}^d} \left\{ \nabla_{\theta} J^{\pi_{\theta}}(\mu)^{\top} \Delta - \frac{1}{2\eta} D_{\text{KL}}(\pi_{\theta_k} \parallel \pi_{\theta_k + \Delta \theta}) \right\}.$$

- ▶ Taylor approximation: $D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\theta + \Delta \theta}) \approx \Delta \theta^{\top} F_{\mu}(\theta) \Delta \theta$ where

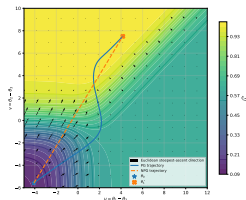
$$F_{\mu}(\theta) := \mathbb{E}_{(s,a) \sim d_{\mu}^{\pi_{\theta}} \otimes \pi_{\theta}} \left[\nabla \log \pi_{\theta}(a|s) \nabla \log \pi_{\theta}(a|s)^{\top} \right]$$

- ▶ It yields Natural Policy Gradient (NPG) [Kak01]

$$\theta_{k+1} = \theta_k + \eta F_{\mu}(\theta)^{-1} \nabla_{\theta} J^{\pi_{\theta}}(\mu).$$

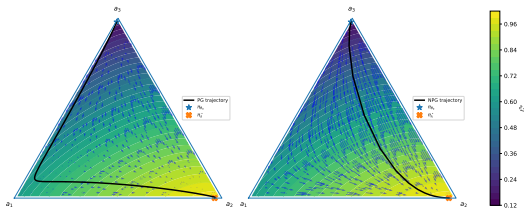
Natural Policy Gradient vs. Policy Gradient in Action

- ▶ Bandit setting: $\mathcal{A} = \{a_1, a_2, a_3\}$ and $\mathcal{S} = \{s\}$
- ▶ Entropy-regularized objective: $J_{\tau}^{\pi_{\theta}} = r^{\top} \pi_{\theta} + \tau H(\pi_{\theta})$.
- ▶ Tabular SoftMax: $\pi_{\theta}(a_i) = \frac{\exp(\theta_i)}{\sum_{j=1}^3 \exp(\theta_j)}$.



In the (Euclidean) parameter space,

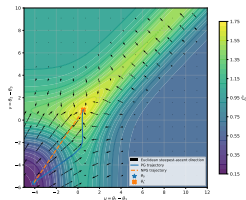
- ▶ PG follows the steepest ascent direction,
- ▶ NPG is preconditioned and has a different direction.



In policy space, NPG is steepest ascent with respect to Fisher geometry.

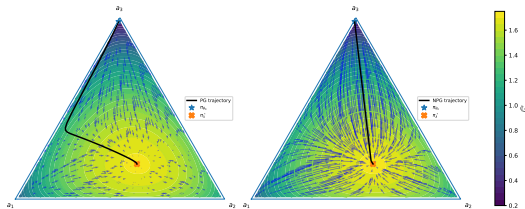
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In the (Euclidean) parameter space,

- ▶ PG follows the steepest ascent direction,
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In policy space, NPG is steepest ascent with respect to Fisher geometry.

Compatible Function Approximation for NPG

Computational challenge: estimate and invert $F_\mu(\theta) \in \mathbb{R}^{d \times d}$ for

$$\theta_{k+1} = \theta_k + \eta F_\mu(\theta)^{-1} \nabla_\theta J^{\pi_\theta}(\mu).$$

Proposition (Compatible Function Approximation, [AKLM21])

Under any policy π_θ , let

$$L_\theta(w) := \mathbb{E}_{(s,a) \sim d_\mu^{\pi_\theta} \otimes \pi_\theta} \left[\left(w^\top \nabla_\theta \log \pi_\theta(a|s) - A^{\pi_\theta}(s,a) \right)^2 \right].$$

Then,

$$F_\mu(\theta)^{-1} \nabla_\theta V^{\pi_\theta}(\mu) = \frac{1}{1-\gamma} \arg \min_{w \in \mathbb{R}^d} L_\theta(w). \quad (\text{CFA})$$

Practical use: CFA turns NPG into an **actor-critic**: fit a compatible critic

$$\hat{A}_w(s,a) = w^\top \nabla_\theta \log \pi_\theta(a|s)$$

(e.g., via TD learning) and update θ via w . No explicit inversion of $F_\mu(\theta)$.

Proof: Just $\nabla_w L_\theta(w) = 0$ (first-order condition for optimality) and (PG-A).

Convergence of Natural Policy Gradient

- ▶ To understand NPG, consider a simplified NPG-Flow ($\eta \downarrow 0$):

$$\begin{aligned} d\theta_t/dt &= F_\mu(\theta_t)^\dagger \nabla_\theta J^{\pi_\theta}(\mu) \\ \theta_0 &= 0. \end{aligned}$$

- ▶ Linear function approximation: $\pi_\theta(a|s) \propto \exp(\theta^\top \phi(s, a))$.

Theorem

Assume

- ▶ *Concentrability:* $\frac{d_{\mu}^{\pi^*}(s)\pi^*(a|s)}{d_{\mu}^{\pi_{\theta_t}}(s)\pi_{\theta_t}(a|s)} \leq C_{\text{conc}} < \infty$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,
- ▶ *Approximation error:*

$$\inf_{w \in \mathbb{R}^d} \mathbb{E}_{(s,a) \sim d_{\mu}^{\pi_{\theta}} \otimes \pi_{\theta}} \left[(\phi(s, a)^\top w - Q^{\pi_{\theta}}(s, a))^2 \right] \leq \epsilon_{\text{app}}$$

Then,

$$\min_{t \in [0, T]} \left(J^{\pi^*}(\mu) - J^{\pi_{\theta_t}}(\mu) \right) \leq \frac{\log |\mathcal{A}|}{(1 - \gamma)T} + \frac{M\epsilon_{\text{app}}}{(1 - \gamma)}$$

Convergence of Natural Policy Gradient

On function approximation error: When does linear approximation work?

- ▶ Consider Linear-MDP:

$$r(s, a) = \phi(s, a)^\top \nu_r \quad \text{and} \quad P(s'|s, a) = \phi(s, a)^\top \nu_P(s').$$

- ▶ For any $\pi \in \Pi$, $Q^\pi(s, a) = \phi(s, a)^\top \nu_\pi^*$ for some $\nu_\pi^* \in \mathbb{R}^d$ (Realizability)
- ▶ Thus, $\epsilon_{\text{app}} = 0$ and $\max_{t \in [0, T]} J^{\pi_{\theta_t}}(\mu) \rightarrow J^{\pi^*}(\mu)$ as $T \rightarrow \infty$.

On exploration:

- ▶ $C_{\text{conc}} < \infty$ is an exploration assumption.
- ▶ Each (s, a) visited by π^* should be visited under π_{θ_t} for all $t \in [0, T]$.
- ▶ Sufficient condition:

$$\pi_{\theta_t}(a|s) \geq p_{\min} > 0 \quad (\text{Persistence of excitation})$$

$$\text{Supp}(d_\mu^{\pi^*}) \subset \text{Supp}(\mu) \quad d_\mu^\pi(s) \geq (1 - \gamma)\mu(s) \text{ for any } \pi.$$

Trust-Region Policy Optimization (TRPO)

Recall the formulation that led to NPG:

$$\theta_{k+1} \in \theta_k + \arg \max_{\Delta\theta \in \mathbb{R}^d} \left\{ \nabla_{\theta} J^{\pi_{\theta_k}}(\mu)^{\top} \Delta\theta - \frac{1}{2\eta} D_{\text{KL}}(\pi_{\theta_k} \parallel \pi_{\theta_k + \Delta\theta}) \right\}.$$

- Impose a hard constraint instead of the penalty:

$$\max_{\theta \in \mathbb{R}^d} \nabla_{\theta} J^{\pi_{\theta_k}}(\mu)^{\top} (\theta - \theta_k) \quad \text{subject to} \quad D_{\text{KL}}(\pi_{\theta_k} \parallel \pi_{\theta}) \leq \delta.$$

- Taylor approximation of $D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\theta + \Delta\theta})$ yields TRPO [SLA⁺15]

$$\max_{\Delta\theta \in \mathbb{R}^d} \nabla_{\theta} J^{\pi_{\theta_k}}(\mu)^{\top} \Delta\theta \quad \text{subject to} \quad \frac{1}{2} \Delta\theta^{\top} F_{\mu}(\theta_k) \Delta\theta \leq \delta.$$

- Solving this via KKT conditions yields

$$\theta_{k+1} = \theta_k + \eta_{\theta_k} F_{\mu}(\theta_k)^{-1} \nabla J^{\pi_{\theta_k}}(\mu) \quad \text{where} \quad \eta_{\theta} := \frac{\sqrt{2\delta}}{\|\nabla J^{\pi_{\theta}}(\mu)\|_{F_{\mu}(\theta)}^{-1}}$$

- TL;DR: Same direction as NPG, but a different step-size η_{θ} .

Entropy Regularization in Policy Optimization

Why Entropy Regularization?

- ▶ Need for exploration: $\pi_{\theta_t}(a|s) > 0$ was assumed.
- ▶ Can we ensure this? Furthermore, can we speed up NPG?

Idea: Entropy regularization. Let

$$H^\pi(\mu) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t h(\pi(\cdot|s)) | s_0 \sim \mu\right], \quad \text{where}$$

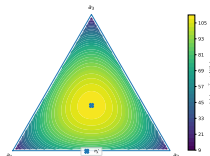
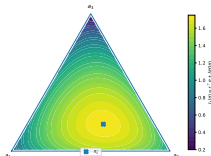
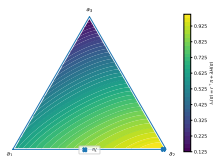
$$h(\pi(\cdot|s)) = - \sum_{a \in \mathcal{A}} \pi(a|s) \log \pi(a|s).$$

For a regularization coefficient $\lambda \geq 0$,

$$J_\lambda^\pi(\mu) := J^\pi(\mu) + \lambda H^\pi(\mu).$$

Why does it work? [\[ALRNS19\]](#)

$$\text{Unif}(\mathcal{A}) \in \arg \max_{\pi \in \Pi} H^\pi(\mu)$$



Entropy Regularization: Structural Results

- **Value functions.** Under any policy π ,

$$V_{\lambda}^{\pi}(s) := \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k \{ r(s_k, a_k) - \lambda \log \pi(a_k | s_k) \} \middle| s_0 = s \right]$$
$$q_{\lambda}^{\pi}(s, a) := r(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\lambda}^{\pi}(s').$$

- **Optimal policy.** $V_{\lambda}^{\star}(s) := \sup_{\pi \in \Pi} V_{\lambda}^{\pi}(s) \rightsquigarrow q_{\lambda}^{\star} = r + \gamma P V_{\lambda}^{\star}$. Then,

$$\pi^{\star}(a|s) = \frac{\exp(q_{\lambda}^{\star}(s, a)/\lambda)}{\sum_{a'} \exp(q_{\lambda}^{\star}(s, a')/\lambda)} \quad \text{is the unique optimal policy.}$$

- **Realizability.** $\Pi_{\Theta} = \{ \pi_{\theta}(a|s) \propto f_{\theta}(s, a) : \theta \in \Theta \}$. If

$$f_{\theta^{\star}} = \frac{1}{\lambda} q_{\lambda}^{\star} \quad \text{for some } \theta^{\star} \in \Theta,$$

then $\pi^{\star} \in \Pi_{\Theta}$.

Entropy Regularization: Policy Gradients

1. Let

$$Q_{\lambda}^{\pi}(s, a) = \mathbb{E} \left[\sum_k \gamma^k \{ r(s_k, a_k) - \lambda \log \pi(a_k | s_k) \} \mid s_0 = s, a_0 = a \right]$$

Then, we have an extended policy gradient theorem:

$$\nabla_{\theta} J_{\lambda}^{\pi_{\theta}}(\mu) = \frac{1}{1 - \gamma} \mathbb{E}_{(s, a) \sim d_{\mu}^{\pi_{\theta}} \otimes \pi_{\theta}} \left[Q_{\lambda}^{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(a | s) \right].$$

2. Linear function approximation $\pi_{\theta}(a | s) \propto \exp(\theta^{\top} \phi(s, a))$:

$$(1 - \gamma) F_{\mu}(\theta)^{-1} \nabla J_{\lambda}^{\pi_{\theta}}(\mu) = -\lambda \theta + \underbrace{\arg \min_{w \in \mathbb{R}^d} \mathbb{E} [|\nabla \log \pi_{\theta}(a | s) w - q_{\lambda}^{\pi_{\theta}}(s, a)|^2]}_{=: g(\theta)}$$

Hence,

$$\theta_{k+1} = (1 - \eta \lambda) \theta_k + \eta g(\theta_k)$$

Entropy regularization induces **weight decay** in Fisher geometry [CHS24].

Entropy-Regularized Natural Policy Gradient: Analysis

For any $\theta \in \mathbb{R}^d$, let $\alpha_\lambda^\pi(s, a) = q_\lambda^\pi(s, a) - \sum_{a'} \pi(a'|s) q_\lambda^\pi(s, a')$ and

$$g(\theta) = \arg \min_{w \in \mathbb{R}^d: \|w\|_2 \leq \rho} \mathbb{E}[|\nabla \log \pi_\theta(a, s)^\top w - \alpha_\lambda^{\pi_\theta}(s, a)|^2],$$

and consider **projected** NPG-Flow:

$$\begin{aligned}\dot{\theta}_t &= g(\theta_t) - \lambda \theta_t, \quad t \geq 0 \\ \theta_0 &= 0.\end{aligned}$$

Key observations.

- Exploration is ensured, not assumed:

$$\sup_{t \geq 0} \|g(\theta_t)\|_2 \leq \rho \quad \Rightarrow \quad \inf_{t \geq 0} \inf_{(s,a) \in \mathcal{S} \times \mathcal{A}} \pi_{\theta_t}(a|s) \geq \frac{\exp(-2\rho/\lambda)}{|\mathcal{A}|} =: p_{\min} > 0$$

- If $q_\lambda^\pi(s, a) = \phi(s, a)^\top w_{\pi, \lambda}$, then

$$\Psi(\pi_{\theta_t}) \leq \exp(-\lambda t) \log |\mathcal{A}|, \quad t \geq 0,$$

$$\text{where } \Psi(\pi) := \mathbb{E}_{d_\mu^{\pi_\lambda^\star} \otimes \pi_\lambda^\star} \left[D_{\text{KL}}(\pi_\lambda^\star(\cdot|s) \parallel \pi(\cdot|s)) \right].$$

- Convergence to **regularized** optimum: $\pi_\lambda^\star \in \arg \max_\pi J_\lambda^\pi(\mu)$.

Entropy Regularization: Main Takeaways

Entropy regularization changes the geometry and dynamics of NPG:

$$J_{\lambda}^{\pi}(\mu) = J^{\pi}(\mu) + \lambda H^{\pi}(\mu).$$

What entropy regularization gives

$$\textbf{Weight decay} \quad : \quad \dot{\theta}_t = g_t - \lambda \theta_t$$

$$\textbf{Uniform exploration} \quad : \quad \|\theta_t\|_2 \leq \rho/\lambda \Rightarrow \pi_{\theta_t}(a|s) \geq \frac{e^{-2\rho/\lambda}}{|\mathcal{A}|}$$

$$\textbf{Lyapunov contraction} \quad : \quad \dot{\Psi}(\pi_{\theta_t}) = -\lambda \Psi(\pi_{\theta_t}) - (1 - \gamma) \Delta_t + \text{ApproxErr}_t$$

where

$$\Psi(\pi_{\theta_t}) = \mathbb{E}_{d_{\mu}^{\pi^*}} D_{\text{KL}}(\pi^*(\cdot|s) \parallel \pi_{\theta_t}(\cdot|s)), \quad \Delta_t = J_{\lambda}^{\pi^*}(\mu) - J_{\lambda}^{\pi_{\theta_t}}(\mu).$$

Exploration is ensured, and convergence is accelerated via weight decay.

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Entropy-Regularized NPG: Lyapunov Drift

Let

$$\Psi(\pi_{\theta_t}) := \mathbb{E}_{s \sim d_{\mu}^{\pi^*}} \left[D_{\text{KL}}(\pi^*(\cdot|s) \parallel \pi_{\theta_t}(\cdot|s)) \right], \quad \Delta_t := J_{\lambda}^{\pi^*}(\mu) - J_{\lambda}^{\pi_{\theta_t}}(\mu).$$

For log-linear SoftMax,

$$\log \pi_{\theta}(a|s) - \mathbb{E}_{\pi_{\theta}} \log \pi_{\theta}(\cdot|s) = \nabla \log \pi_{\theta}(a|s)^{\top} \theta.$$

Hence, with

$$\alpha_{\lambda}^{\pi}(s, a) := q_{\lambda}^{\pi}(s, a) - \mathbb{E}_{\pi(\cdot|s)} q_{\lambda}^{\pi}(s, \cdot),$$

the soft advantage satisfies

$$A_{\lambda}^{\pi_{\theta}}(s, a) = \alpha_{\lambda}^{\pi_{\theta}}(s, a) - \lambda \nabla \log \pi_{\theta}(a|s)^{\top} \theta.$$

If $g_t := g(\theta_t)$ and

$$\dot{\theta}_t = g_t - \lambda \theta_t, \quad \varepsilon_t(s, a) := \nabla \log \pi_{\theta_t}(a|s)^{\top} g_t - \alpha_{\lambda}^{\pi_{\theta_t}}(s, a),$$

then

$$\nabla \log \pi_{\theta_t}(a|s)^{\top} \dot{\theta}_t = A_{\lambda}^{\pi_{\theta_t}}(s, a) + \varepsilon_t(s, a).$$

Therefore, by the entropy-regularized performance difference lemma,

$$\dot{\Psi}(\pi_{\theta_t}) = -\lambda \Psi(\pi_{\theta_t}) - (1 - \gamma) \Delta_t - \mathbb{E}_{d_{\mu}^{\pi^*} \otimes \pi^*} [\varepsilon_t(s, a)].$$

In particular, if $\varepsilon_t \equiv 0$, then

$$\dot{\Psi}(\pi_{\theta_t}) \leq -\lambda \Psi(\pi_{\theta_t}), \quad \Psi(\pi_{\theta_t}) \leq e^{-\lambda t} \log |\mathcal{A}|.$$

Entropy-Regularized NPG: Approximation Error

Recall

$$\dot{\Psi}(\pi_{\theta_t}) = -\lambda \Psi(\pi_{\theta_t}) - (1 - \gamma) \Delta_t - \mathbb{E}_\star[\varepsilon_t], \quad \mathbb{E}_\star[\cdot] := \mathbb{E}_{d_\mu^{\pi^\star} \otimes \pi^\star}[\cdot].$$

The residual is controlled under the on-policy distribution:

$$\varepsilon_t(s, a) = \nabla \log \pi_{\theta_t}(a|s)^\top g_t - \alpha_\lambda^{\pi_{\theta_t}}(s, a),$$
$$g_t \in \arg \min_{\|w\|_2 \leq \rho} \mathbb{E}_{d_\mu^{\pi_t} \otimes \pi_t} \left[\left(\nabla \log \pi_t(a|s)^\top w - \alpha_\lambda^{\pi_t}(s, a) \right)^2 \right].$$

By change of measure and Cauchy–Schwarz,

$$\left| \mathbb{E}_\star[\varepsilon_t] \right| \leq \underbrace{\left\| \frac{d_\mu^{\pi^\star} \pi^\star}{d_\mu^{\pi_t} \pi_t} \right\|_{L^2(d_\mu^{\pi_t} \pi_t)}}_{\text{distribution shift}} \cdot \underbrace{\left(\mathbb{E}_{d_\mu^{\pi_t} \otimes \pi_t}[\varepsilon_t^2] \right)^{1/2}}_{\text{compatible approx. error}} =: \mathcal{E}_t.$$

Hence

$$\dot{\Psi}(\pi_{\theta_t}) \leq -\lambda \Psi(\pi_{\theta_t}) - (1 - \gamma) \Delta_t + \mathcal{E}_t.$$

If $\mathcal{E}_t \leq \mathcal{E}$, then integrating over $[0, T]$ gives

$$\boxed{\frac{1}{T} \int_0^T \Delta_t dt \leq \frac{\log |\mathcal{A}|}{(1 - \gamma)T} + \frac{\mathcal{E}}{1 - \gamma}.}$$

Entropy-Regularized Performance Difference Lemma

Lemma

For any policies π, π' ,

$$J_{\lambda}^{\pi}(\mu) - J_{\lambda}^{\pi'}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{d_{\mu}^{\pi} \otimes \pi} \left[A_{\lambda}^{\pi'}(s, a) + \lambda \log \frac{\pi'(a|s)}{\pi(a|s)} \right],$$

where

$$A_{\lambda}^{\pi'}(s, a) = Q_{\lambda}^{\pi'}(s, a) - V_{\lambda}^{\pi'}(s).$$

Proof. Using $q_{\lambda}^{\pi'}(s, a) = r(s, a) + \gamma \mathbb{E}[V_{\lambda}^{\pi'}(s')|s, a]$, we have

$$r - \lambda \log \pi = A_{\lambda}^{\pi'} + \lambda \log \frac{\pi'}{\pi} + V_{\lambda}^{\pi'} - \gamma P V_{\lambda}^{\pi'}.$$

Hence, along trajectories of π ,

$$J_{\lambda}^{\pi}(\mu) = \mathbb{E}_{\pi} \sum_{t \geq 0} \gamma^t \left[A_{\lambda}^{\pi'}(s_t, a_t) + \lambda \log \frac{\pi'(a_t|s_t)}{\pi(a_t|s_t)} + V_{\lambda}^{\pi'}(s_t) - \gamma V_{\lambda}^{\pi'}(s_{t+1}) \right].$$

The last two terms telescope to $J_{\lambda}^{\pi'}(\mu)$. Thus

$$J_{\lambda}^{\pi}(\mu) - J_{\lambda}^{\pi'}(\mu) = \mathbb{E}_{\pi} \sum_{t \geq 0} \gamma^t \left[A_{\lambda}^{\pi'}(s_t, a_t) + \lambda \log \frac{\pi'(a_t|s_t)}{\pi(a_t|s_t)} \right].$$